Taoyuan Wang

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Experience

Software Engineer at Wing (Alphabet) Oct 2022 – Present
UAS (Uncrewed Aircraft Systems) Traffic Management (UTM) Team *Palo Alto, CA*

- Main backend engineer for the human-in-the-loop supervision system behind FAA approval for a **10× increase in aircraft per operator**, shifting operators to management by exception
- Built C++ services for 4D (3D + time) spatial constraint checking with **S2 Geometry**; precomputed geometric joins offline in a Flume pipeline
- Led design (eng lead) of config-driven RBAC with per-RPC permission checks; owned the pilot License service and its dual-write Spanner migration without a maintenance window
- Led end-to-end validation of a safety-critical emergency-landing command on **simulated vehicles** in pre-production, asserting state transitions against deadlines; wrote the on-call debugging playbook
- Independently rebuilt a multi-drone delivery PoC in MuJoCo (simulation): **Gemini Robotics-ER** perception from camera pixels with depth back-projection and Kalman tracking, plus a geometric SE(3) flight controller
- Built a Simplex safety layer and evaluation benchmark: deterministic checks gate every Gemini plan; an independent 50 Hz ground-truth auditor scores runs (**0 violations** in 300 adversarial plans + 240 runs)
- Built PoC Gemini-in-the-loop operator tools (models propose, deterministic checks + humans decide): schema-constrained 4D flight-restriction drafting and incident triage tested via time-cutoff replay

Software Engineer at Appen (fka Figure Eight) Oct 2021 – Oct 2022
Computer Vision Tools Team (Internal transfer from China) *Sunnyvale, CA*

- Led end-to-end development of ML-assisted, human-verified **3D/LiDAR** annotation tools (React/Three.js, Flask/Spring): click-to-box via DBSCAN on k-d trees, and GPU shaders + octree culling (1M to 10M+ points)
- Extended cuboid annotation into 4D (3D + time) point-cloud semantic segmentation as a PoC side project, later launched as a product used by **20+ leading clients** in autonomous vehicles and robotics

Software Engineer at Appen China Apr 2019 – Sep 2021
Computer Vision Tools Team *Shanghai, China*

- Founding engineer at Appen China; built the localized annotation platform for the Chinese market
- Prototyped 3D point-cloud object detection/tracking for pre-labeling and NumPy LiDAR-to-camera projection from device calibration intrinsics/extrinsics to support multi-sensor fusion annotation
- Built annotator event logging and workflow analytics with UX designers and data scientists, driving training and tooling improvements that reduced labeling costs by **56%**

Software Engineer at Works Applications Apr 2018 – Mar 2019
AI WORKS HUE Tech Lead Group *Shanghai, China*

- Built full-stack enterprise ERP features (Vue.js frontend, Spring/Hibernate backend) and prototyped a transformer-based chatbot and LSTM-based OCR (Tesseract) for workflow automation

Research Assistant at Academia Sinica Dec 2017 – Mar 2018
Natural Language Processing Lab *Taipei, Taiwan*

- Researched conversational agents for Chinese short text and maintained a large Weibo post–comment dataset
- Implemented Seq2Seq with attention in PyTorch, fine-tuned with **reinforcement learning** using human-rated comment appropriateness scores as rewards (an early RLHF-style setup)

IT Specialist at Social Affairs Bureau (Military Service) Nov 2016 – Oct 2017

Education

National Tsing Hua University Hsinchu, Taiwan
M.S. in Computer Science (Graduate Researcher at Internet Lab) *Sep 2015 – Aug 2016*
B.S. in Computer Science (Exchanged to Beijing Tsinghua and Nagoya University) *Sep 2010 – Aug 2015*

Skills

Languages: C++, Python, Java, TypeScript/JavaScript
Backend & Data: gRPC, Protocol Buffers, Spanner, Flume, Docker, GCP, AWS, Flask, Spring
Robotics & ML: MuJoCo, Gemini Robotics-ER, Gemini API, PyTorch, NumPy, S2 Geometry, Three.js (WebGL)
Tools & Certification: pytest, JUnit, Jest, Git, Jenkins, Google Antigravity; Google C++ Readability (2024)